

DIFFERENTIATION AND SUPPORT

A science fair unit reaches students with very different starting points. These strategies keep the science rigorous while removing barriers that have nothing to do with ability.

STUDENTS WHO NEED MORE STRUCTURE

- Assign from the Grades 5 to 6 experiment menu regardless of grade level. A guided experiment done thoroughly beats an open topic abandoned in week six.
- Break each milestone into two smaller due dates and check in at both.
- Provide a pre-formatted data table rather than asking the student to design one.
- Pair the student with a partner for data collection while keeping the analysis and write-up individual.

STUDENTS READY FOR MORE CHALLENGE

- Require a second independent variable tested in a separate trial set.
- Ask for a larger sample size and a discussion of why sample size affects confidence.
- Require a primary source: an interview with a researcher or a professional in the field.
- Point them toward OAS regional entry and the FairGame mentor matching program.

MULTILINGUAL LEARNERS

- Allow the journal to be kept in the student's first language, with the final report in English.
- Pre-teach the ten vocabulary terms for each phase rather than all forty at once.
- Accept labeled diagrams in place of written descriptions during the design phase.
- Provide sentence frames for the CER conclusion: my claim is, my evidence is, this matters because.

STUDENTS WITH LIMITED HOME SUPPORT

- Keep all work at school. The Grades 5 to 6 model is built for exactly this and works at any grade.
- Provide all materials from the classroom supply. Never require a purchase.
- Schedule data collection during class rather than as homework.
- Offer a lunch or study hall work session as the equivalent of home project time.

DIFFERENTIATION AND SUPPORT, CONTINUED

STUDENTS WITH IEPs OR 504 PLANS

- Apply existing accommodations directly: extended time on milestones, scribing for the journal, oral instead of written responses where the plan allows.
- Allow the presentation to be delivered to one adult rather than a group where anxiety is documented.
- Use the board template as a fill-in structure rather than an open design task.

THE EQUITY PRINCIPLE BEHIND ALL OF THIS

Students in under-resourced schools do not lack intelligence or curiosity. They lack infrastructure and connections. Every choice above removes a barrier without lowering the science.